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Multimodal vortex-induced vibration mitigation and design approach of bistable nonlinear energy sink inerter on bridge structure

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ABSTRACT

Large-scale structures, e.g., long-span bridge structures, are prone to induce multi-modal vibrations due to their densely spaced low modal frequencies. Due to the limited frequency bandwidth of linear dynamic absorbers, they are incapable of effectively mitigating vibrations across multiple modes. To this end, the bistable nonlinear energy sink inerter (BNESI) is used to mitigate the multimodal vortex-induced vibration (VIV) of the beam structure. The highly nonlinear equilibrium differential equations of the beam-BNESI system are numerically solved, and the simulated annealing (SA) algorithm is employed to determine the optimal VIV reduction ratio and BNESI parameters. In comparison to the cubic-type nonlinear energy sink inerter (CNESI), BNESI is found to possess more stable equilibrium positions, smaller stiffness coefficients, and higher VIV mitigation efficiency. The selection of design modes has been found to influence the efficiency of multimodal VIV mitigation, with the use of the intermediate modal order as the design mode resulting in the highest efficiency for multimodal VIV mitigation. The performance-based multimodal VIV mitigation design can be realized with three parameters, i.e., inertance ratio, damping coefficient, and stiffness coefficient. Moreover, the performance-based multimodal VIV mitigation approach and models proposed in this study demonstrate a high level of precision.

1. Introduction

Large-scale structures such as high-rise buildings and bridge structures, due to their significant flexibility characteristics, are susceptible to vibration that poses risks to structural integrity and human safety under the influence of seismic events, wind loads, and other external extreme conditions. Recently, wind-induced vibration issues in bridge structures have been widely recognized, attributed to the significant wind sensitivity resulting from increased spans. The typical wind-induced vibrations of bridge structures include flutter (Zhang et al., 2021), galloping (Yu et al., 2022a), buffeting (Cui et al., 2022), and vortex-induced vibration (VIV) (Ge et al., 2022). These vibrations generally can be mitigated or eliminated through modifications to the aerodynamic shape of the bridge, known as aerodynamic countermeasures (Wang et al., 2023; Zhao et al., 2022). To determine effective aerodynamic measures, it is common practice to conduct physical and numerical wind tunnel tests, systematically adjusting and selecting suitable measures based on the experimental findings (Gao et al., 2021; Wang et al., 2021a). Nonetheless, the effectiveness of aerodynamic measures cannot be guaranteed due to variations between the actual bridge wind environment and test conditions. Moreover, aerodynamic measures do not contribute to improving the structural modal damping. This presents a greater challenge in suppressing damping-sensitive wind-induced vibrations, such as VIV (Xu et al., 2018).

Energy dissipators and passive dynamic vibration absorbers are regarded as mechanical control measures capable of significantly enhancing the structural modal damping ratio, and they are widely utilized in structural vibration mitigation (Koutsoloukas et al., 2022; Sun et al., 2022). Among these, tuned mass dampers (TMDs) are regarded as an effective strategy for addressing VIV in bridge structures, owing to their simple construction and high efficiency in vibration mitigation. Specifically, TMDs have been integrated into the infrastructure of Denmark's Great East Belt Bridge (Larose et al., 1995), Japan's Trans-Tokyo Bay Bridge (Fujino and Abé, 1993), Brazil's Rio-Niteroi Bridge (Battista and Pfeil, 2000), and China's Chong Qi

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Nomenclature

A	pth-order normalized modal coordinate
$a_r(t)$	rth-order modal coordinate
BNESI	Bistable nonlinear energy sink inerter
b	Inertance, kg
CNESI	Cubic-type nonlinear energy sink inerter
C_B	Damping coefficient of BNESI, N·s/m
C_s	Damping coefficient of beam, N·s/m
D	Characteristic length of beam, m
E	Elastic modulus, Pa
f	Nonlinear stiffness force of BNESI, N
$\widetilde{f}$	Approximated nonlinear stiffness force of BNESI, N
$f_K(t)$	Total stiffness force of BNESI, N
$f_C(t)$	Damping force of BNESI, N
$f_I(t)$	Inertial force of BNESI, N
$F(x,t)$	External load, N
$F_K(t)$	Total stiffness force of beam, N
$F_c(t)$	Damping force of beam, N
$F_I(t)$	Inertial force of beam, N
I	Moment of inertia, m^4
k_1	Linear stiffness coefficient, N/m
k_2	Cubic stiffness coefficient, N/m^3
k_3	Fifth order stiffness coefficient, N/m^5
k_4	Seventh order stiffness coefficient, N/m^7
k_{n1}	Dimensionless linear stiffness coefficient
k_{n2}	Dimensionless cubic stiffness coefficient
k_{n3}	Dimensionless fifth order stiffness coefficient
k_{n4}	Dimensionless seventh order stiffness coefficient
m_B	Mass of BNESI
m_p	pth-order modal mass, kg
m_s	Mass per unit length of beam
m_r	Reduced mass ratio
$MVCI$	Multimodal vibration control index
R_i	VIV reduce ratio of each mode
S	Initial horizontal deviation distance of oscillator, m
s	Dimensionless initial horizontal deviation distance of oscillator
U	Wind velocity, m/s
U_r	Reduced wind speed
v	Dynamic displacement of BNESI, m
V	Dimensionless displacement of nonlinear oscillator
$y(x,t)$	Dynamic displacement, m
$y(x_d,t)$	Dynamic displacement of the mass block, m
$y(x_e,t)$	Dynamic displacement of the inerter installation position on beam, m
Y_1, ε	Aeroelastic damping coefficients of VIV
$\overline{z}$	Relative displacement of nonlinear oscillator, m
z	Relative displacement between BNESI and the beam, m
β	Inertance ratio of BNESI
Δ_ϕ	Modal difference
$\phi_r(x)$	rth-order mode shape of beam
$\phi_r(d)$	Value of mode shape at BNESI position
$\phi_r(e)$	Value of mode shape at inerter coupling beam position
ξ_p	pth-order modal damping ratio
λ	Damping coefficient of BNESI
μ	Mass ratio of BNESI
ρ	Air density, kg/m^3
τ	Dimensionless time
ω_p	pth-order modal angular frequency, rad/s
χ	Dimensionless spanwise coordinate

Bridge (Meinhardt et al., 2014), effectively addressing VIV concerns and yielding commendable control achievements. Long-span bridges, due to their high sensitivity to wind, exhibit characteristics of low modal frequency and multi-mode vibration in VIV (Li et al., 2011; Li et al., 2014). In such situations, TMDs have shown limited effectiveness in suppressing VIV in long-span bridges, due to their excessive static displacement at low modal frequencies, e.g., $\Delta > 6$ m when $f < 0.2$ Hz, and narrow frequency bandwidth. The dual-end device proposed by Smith (2002), known as an inerter and characterized by mass amplification effects, has been found to significantly reduce static displacements at low modal frequencies when combined with a TMD (Dai et al., 2021; Xu et al., 2020). Nonetheless, the tuned mass damper inerter (TMDI) retains the narrow bandwidth property inherent in TMD, restricting its capacity to mitigate vibrations to a single mode, and exhibiting low control robustness. The conventional strategy involves the utilization of distributed tuned mass dampers (DTMDs) (Igusa and Xu, 1994; Peng et al., 2022) and multiple tuned mass dampers (MTMDs) (Debnath et al., 2016; Xu et al., 2023) to respectively enhance the robustness of linear dynamic absorbers and achieve multi-mode vibration control. In particular, the former approach employs the arrangement of multiple tuned mass dampers with varying frequencies to control a single-mode vibration, thus augmenting the robustness of control. Meanwhile, the latter method entails the deployment of multiple TMDs with diverse frequencies to regulate vibrations across multiple modes, thereby achieving multi-mode vibration mitigation. The efficacy of this strategy in vibration suppression is low, often requiring extensive optimization processes to achieve desired outcomes. Ideally, an effective control approach should enable broad-frequency multi-mode vibration suppression with a single controller.

A nonlinear dynamic absorber termed a nonlinear energy sink (NES), has been found to possess wideband vibration absorption capabilities through targeted energy transfer (TET), allowing for transient resonance capture (TRC) with different frequency ratios (Gendelman et al., 2001; Vakakis and Gendelman, 2001). The most extensively studied NES employs cubic springs, wherein the restoring force is cubically proportional to the relative displacement between the NES and the primary structure. The control effectiveness of cubic-type NESs (CNESs) has been validated through numerous theoretical and experimental investigations (Vakakis et al., 2008). However, existing research indicates that CNESs exhibit an initial energy threshold. Only when the external excitation energy surpasses this energy threshold does this CNES achieve wideband vibration attenuation (Quinn et al., 2008; Sapsis et al., 2009). This is attributed to the fact that if the input energy is at a lower level, the tangent stiffness of the CNES remains within a smaller range, and the resonance frequency of the CNES is much lower than the natural frequency of the primary structure (Wang et al., 2021b). In such cases, the vibration mitigation performance of the CNES deteriorates significantly. To address the drawbacks above of CNESs, a novel type of NES comprising negative linear stiffness and higher-order nonlinear stiffness terms, known as the bistable NES (BNES) (Al-Shudeifat, 2014), has been demonstrated to possess superior vibration mitigation performance compared to conventional CNESs under input low-energy (Al-Shudeifat and Saeed, 2021; Chen et al., 2021; Fang et al., 2016; Romeo et al., 2015).

Similar to linear vibration absorbers, the vibration control performance of NES is significantly affected by the mass ratio. Typically, a higher mass ratio leads to superior vibration control effectiveness (Parseh et al., 2015). However, for large-scale structures, increasing the mass ratio of auxiliary components is not straightforward. This challenge is particularly evident in bridge structures, where spatial limitations within the box girder constrain the ability to significantly improve the performance of dynamic vibration absorbers. Inspired by the combination of TMD and inerters, recent advancements have integrated inerters with NES to create a nonlinear energy sink inerter (NESI) system (Zhang et al., 2019a; Zhang et al., 2019b). This integration aims to achieve mass amplification within the control system. Compared to traditional NES, NESI requires a smaller mass to achieve the same level of vibration suppression. Moreover, for an equivalent mass, NESI offers enhanced vibration mitigation performance. To date, NESI has demonstrated remarkable broadband vibration absorption capabilities and robustness under different external influences such as earthquakes (Wang et al., 2022), wind loads (Yu et al., 2022b), and pedestrian loads (Wang and Jun, 2022b). While research on NESI's impact on structural vibration control is in its early stages, studies specifically addressing wind-induced vibration response control in bridge structures remain relatively scarce.

Long-span bridge structures are frequently reported to experience multimodal VIV under wind action due to the presence of closely spaced low-modal frequencies. To address this issue more effectively, this study proposes an efficient bistable energy sink inerter (BNESI) and conducts analyses of the multimodal vibration mitigation performance of the beam-BNESI system. The two-terminal inerter is connected to the BNESI mass block and the primary structure, respectively. The equilibrium differential equation of the beam-BNESI system is established and derived, resulting in universal dynamic response equations for the beam-BNESI system. The multimodal vibration mitigation characteristics of BNESI are discussed and the optimal multimodal vibration control index (*MVCI*) corresponding to various parameters is determined by simulated annealing (SA) algorithm. To facilitate the design of BNESI in civil engineering applications, the parameter sensitivity of BNESI is analyzed, and non-sensitive parameters are identified. The performance-based design approach and model are proposed to mitigate the multimodal VIV vibration and have been demonstrated to possess good accuracy.

2. Dynamic system of beam coupled with BNESI

2.1. BNESI system

Fig. 1 depicts the physical arrangement of the BNESI. Within this system, the length of spring in the absence of tension or compression is represented by l, whereas at the unstable equilibrium point ($z = S$), the vertical length is denoted as l_0. The inerter serving as an ancillary component is connected to the mass block and primary structure, with its apparent mass denoted as b. The relationship between the spring length and the stable equilibrium position is expressed by the following formula:

$$l = \sqrt{S^2 + {l_0}^2} \tag{1}$$

The derivation of the nonlinear force exerted by the springs on the mass is as follows:

$$f = -2k(z - S)\left(1 - l\left({l_0}^2 + (z - S)^2\right)^{-\frac{1}{2}}\right) \tag{2}$$

The application of Taylor series expansion to the nonlinear force formulation facilitates an approximation of the nonlinear force. This derived expression has been shown to satisfy accuracy requirements, as evidenced by the findings of Al-Shudeifat (2014):

$$\widetilde{f} = k_1\overline{z} - k_2\overline{z}^3 + k_3\overline{z}^5 - k_4\overline{z}^7 \tag{3}$$

let $\widetilde{f}$ represent the approximated nonlinear force expanded using the Taylor series, where $\overline{z} = z - S$ denotes the relative displacement between the primary structure and BNES. Here, z signifies the relative displacement between the primary structure and BNES, while S indicates the initial horizontal deviation distance of the nonlinear oscillator. The nonlinear stiffness coefficients are expressed as follows: $k_1 = 2k\left(\frac{l}{l_0} - 1\right)$, $k_2 = \frac{kl}{{l_0}^3}$, $k_3 = \frac{3kl}{4{l_0}^5}$, and $k_4 = \frac{5kl}{8{l_0}^7}$. Upon introducing dimensionless variables $\alpha = \frac{l_0}{l}$ and $\gamma = \frac{l}{D}$, the following relationships emerge:

$$k_1 = \frac{2k(1-\alpha)}{\alpha}, k_2 = \frac{k}{\alpha^3\gamma^2D^2}, k_3 = \frac{3k}{4\alpha^5\gamma^4D^4}, k_4 = \frac{5k}{8\alpha^7\gamma^6D^6} \tag{4}$$

For the given parameters $k = 1N/m$, $l = 1m$, and $l_0 = 0.97m$, the nonlinear stiffness force of bistable nonlinear energy sink is compared with that of cube nonlinear energy sink. The nonlinear stiffness forces versus displacement are depicted in Fig. 2. Due to the initial linear negative stiffness, the bistable nonlinear energy sink is observed to possess two stable equilibrium positions and one unstable equilibrium position, while the cubic nonlinear energy sink only has one stable equilibrium point. Hence, under low inputted energy, the BNES can

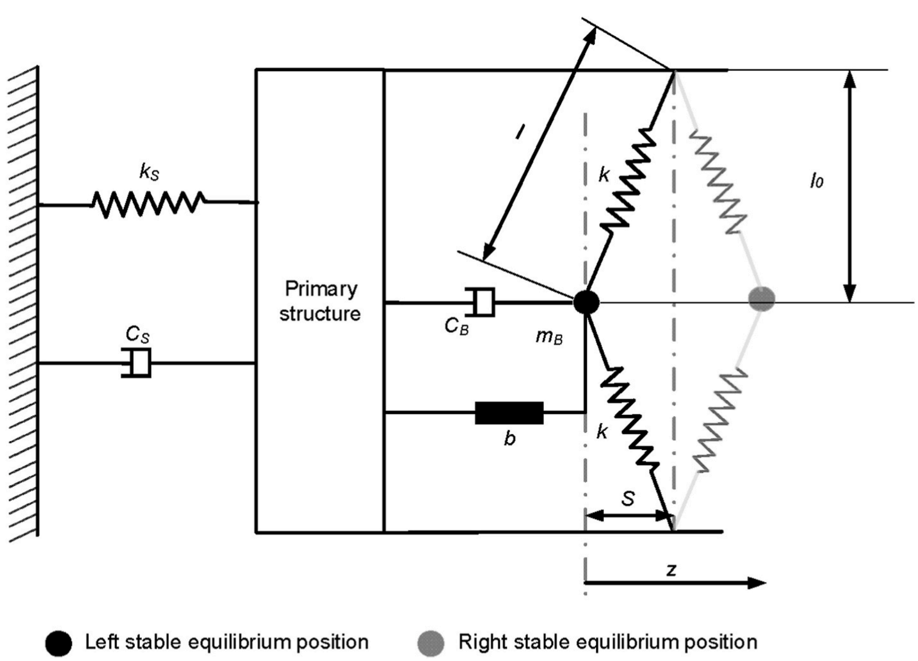


Fig. 1. Physical configuration of BNESI.

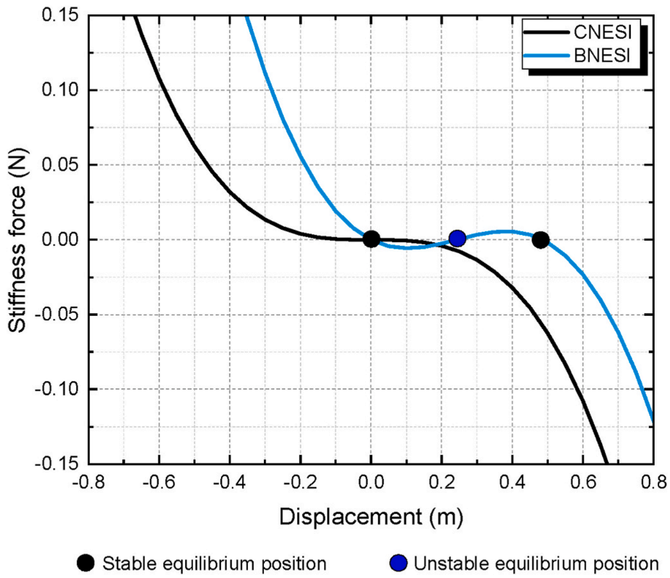


Fig. 2. Nonlinear stiffness forces versus displacement of nonlinear energy sink.

operate normally, while the NES cannot mitigate vibration of the primary structure.

2.2. Beam coupled with BNESI system

Within the framework of the BNESI-integrated beam system, the dynamic displacement $y(x,t)$ of the beam structure, incorporating the BNESI, is a consequence of the influence of external loads $F(x,t)$ during the vibration process. These loads are countered by a composite of inherent stiffness, damping, and inertia components within the primary structure (beam), along with contributions from the BNESI. For the uniform Euler-Bernoulli beam, these amalgamated components are delineated as follows:

$$F_K(t) = EI\frac{\partial^4 y(x,t)}{\partial x^4} \tag{5}$$

$$F_c(t) = C_s\frac{\partial y(x,t)}{\partial t} \tag{6}$$

$$F_I(t) = m_s\frac{\partial^2 y(x,t)}{\partial t^2} \tag{7}$$

where $F_K(t)$, $F_c(t)$, and $F_I(t)$ are total stiffness force, damping force, and inertial force of beam respectively; E is the elastic modulus; I is the moment of inertia; C_s is the viscous damping coefficient; m_s is mass per unit length of beam.

The cumulative nodal force upon the installation of BNESI can be formally defined as:

$$f_K(t) = k_1\bar{z} - k_2\bar{z}^3 + k_3\bar{z}^5 - k_4\bar{z}^7 \tag{8}$$

$$f_C(t) = C_B[\dot{y}(x_d,t) - \dot{v}(t)] \tag{9}$$

$$f_I(t) = b[\ddot{y}(x_e,t) - \ddot{v}(t)] \tag{10}$$

where $f_K(t)$, $f_c(t)$, and $f_I(t)$ are total stiffness force, damping force, and inertial force of BNESI respectively; the relative displacement $(\bar{z})$ of nonlinear oscillator is presented as $\bar{z} = z - S$; z denotes the relative displacement between BNESI and the beam, defined as $z = v - y$; here, $y(x_e,t)$ represents the dynamic displacement of the inerter installation position on the beam, with the overdot indicating the derivative with respect to time t; the damping coefficient of BNESI is denoted as C_B; b signifies the inertance (apparent mass), surpassing its physical mass.

At the vibration instant $T = t$, the coupled system maintains the ensuing equilibrium condition:

$$F_k(t) + F_C(t) + F_I(t) + [f_K(t) + f_C(t)]\delta(x - x_d) + f_I(t)\delta(x - x_e) = F(x,t) \tag{11}$$

$$m_B\ddot{v}(t) - f_K(t) - f_C(t) - f_I(t) = 0 \tag{12}$$

The dynamic displacement can be decomposed into the summation of modal displacements as:

$$y = \sum_{r=1}^{\infty}\phi_r(x)a_r(t) \tag{13}$$

where $\phi_r(x)$ is the rth-order mode shape of the beam; $a_r(t)$ is the rth-order modal coordinate of the beam. Substituting Eqs. (5) ~(10) and Eq. (13) into Eqs. (11) and (12) leads to

$$\begin{aligned} &EI\sum_{r=1}^{\infty}\frac{d^4\phi_r(x)}{dx^4}a_r(t) + C_s\sum_{r=1}^{\infty}\phi_r(x)\dot{a}_r(t) + m_s\sum_{r=1}^{\infty}\phi_r(x)\ddot{a}_r(t) \\ &+ \left\{k_1\left[v(t) - \sum_{r=1}^{\infty}\phi_r(d)a_r(t) - S\right]\right. \\ &- k_2\left[v(t) - \sum_{r=1}^{\infty}\phi_r(d)a_r(t) - S\right]^3 + k_3\left[v(t) - \sum_{r=1}^{\infty}\phi_r(d)a_r(t) - S\right]^5 \\ &\left.- k_4\left[v(t) - \sum_{r=1}^{\infty}\phi_r(d)a_r(t) - S\right]^7 - C_B\left[\dot{v}(t) - \sum_{r=1}^{\infty}\phi_r(d)\dot{a}_r(t)\right]\right\}\delta(x-d) \\ &- b\left[\ddot{v}(t) - \sum_{r=1}^{\infty}\phi_r(e)\ddot{a}_r(t)\right]\delta(x-e) = F(x,t) \end{aligned} \tag{14}$$

$$\begin{aligned} &m_B\ddot{v}(t) - k_1\left[v(t) - \sum_{r=1}^{\infty}\phi_r(d)a_r(t) - S\right] + k_2\left[v(t) - \sum_{r=1}^{\infty}\phi_r(d)a_r(t) - S\right]^3 \\ &- k_3\left[v(t) - \sum_{r=1}^{\infty}\phi_r(d)a_r(t) - S\right]^5 + k_4\left[v(t) - \sum_{r=1}^{\infty}\phi_r(d)a_r(t) - S\right]^7 \\ &+ C_B\left[\dot{v}(t) - \sum_{r=1}^{\infty}\phi_r(d)\dot{a}_r(t)\right] + b\left[\ddot{v}(t) - \sum_{r=1}^{\infty}\phi_r(e)\ddot{a}_r(t)\right] = 0 \end{aligned} \tag{15}$$

where $\phi_r(d)$ presents the value of mode shape at BNESI position; $\phi_r(e)$ is the value of mode shape at inerter coupling beam position. Since these modes are solutions to a Sturm–Liouville eigenvalue problem they satisfy the following orthonormality relations:

$$\int_0^L \phi_r(x)\phi_p(x)dx = 0, r \neq p \tag{16}$$

$$\int_0^L m_s\phi_r(x)\phi_p(x)dx = m_p\delta_{rp}, r,p = 1,2,\ldots \tag{17}$$

$$\int_0^L EI\frac{\partial^2\phi_r}{\partial x^2}\frac{\partial^2\phi_p}{\partial x^2}dx = m_p\omega_p^2\delta_{rp}, r,p = 1,2,\ldots \tag{18}$$

Upon multiplication of Eqs. (14) and (15) by an arbitrary mode shape $\phi_p(x)$, and subsequent integration over x from 0 to L:

$$\begin{aligned}&m_p\left[\omega_p^2a_p(t)+2\omega_p\xi_p\dot{a}_p(t)+\ddot{a}_p(t)\right]+\left\{k1\left[v(t)-\sum_{r=1}^{\infty}\phi_r(d)a_r(t)-S\right]\right.\\&-k_2\left[v(t)-\sum_{r=1}^{\infty}\phi_r(d)a_r(t)-S\right]^3+k_3\left[v(t)-\sum_{r=1}^{\infty}\phi_r(d)a_r(t)-S\right]^5\\&\left.-k_4\left[v(t)-\sum_{r=1}^{\infty}\phi_r(d)a_r(t)-S\right]^7-C_B\left[\dot{v}(t)-\sum_{r=1}^{\infty}\phi_r(d)\dot{a}_r(t)\right]\right\}\phi_p(d)\\&-b\left[\ddot{v}(t)-\sum_{r=1}^{\infty}\phi_r(e)\ddot{a}_r(t)\right]\phi_p(e)=\int_0^L F(x,t)\phi_p(x)\end{aligned}\tag{19}$$

$$\begin{aligned}&m_B\ddot{v}(t)-k_1\left[v(t)-\sum_{r=1}^{\infty}\phi_r(d)a_r(t)-S\right]+k_2\left[v(t)-\sum_{r=1}^{\infty}\phi_r(d)a_r(t)-S\right]^3\\&-k_3\left[v(t)-\sum_{r=1}^{\infty}\phi_r(d)a_r(t)-S\right]^5+k_4\left[v(t)-\sum_{r=1}^{\infty}\phi_r(d)a_r(t)-S\right]^7\\&+C_B\left[v(t)-\sum_{r=1}^{\infty}\phi_r(d)\dot{a}_r(t)\right]+b\left[\ddot{v}(t)-\sum_{r=1}^{\infty}\phi_r(e)\ddot{a}_r(t)\right]=0\end{aligned}\tag{20}$$

where m_p is the pth-order modal mass; ω_p is the pth-order modal angular frequency; ξ_p is the pth-order modal damping ratio.

In the scenario where the primary structure is predominantly influenced by a single mode, denoted as $y(x,t)=\phi_p(d)a_p(t)$, the equations of motion for the beam coupled with the BNESI system can be streamlined as follow

$$\begin{aligned}&m_p\left[\omega_p^2a_p(t)+2\omega_p\xi_p\dot{a}_p(t)+\ddot{a}_p(t)\right]+\{k1\left[v(t)-\phi_p(d)a_p(t)-S\right]\\&-k_2\left[v(t)-\phi_p(d)a_p(t)-S\right]^3+k_3\left[v(t)-\phi_p(d)a_p(t)-S\right]^5\\&-k_4\left[v(t)-\phi_p(d)a_p(t)-S\right]^7-C_B\left[\dot{v}(t)-\phi_p(d)\dot{a}_p(t)\right]\}\phi_p(d)\\&-b\left[\ddot{v}(t)-\phi_p(e)\ddot{a}_p(t)\right]\phi_p(e)=\int_0^L F(x,t)\phi_p(x)\end{aligned}\tag{21}$$

$$\begin{aligned}&m_B\ddot{v}(t)-k_1\left[v(t)-\phi_p(d)a_p(t)-S\right]+k_2\left[v(t)-\phi_p(d)a_p(t)-S\right]^3\\&-k_3\left[v(t)-\phi_p(d)a_p(t)-S\right]^5+k_4\left[v(t)-\phi_p(d)a_p(t)-S\right]^7+C_B\left[\dot{v}(t)\right.\\&\left.-\phi_p(d)\dot{a}_p(t)\right]+b\left[\ddot{v}(t)-\phi_p(e)\ddot{a}_p(t)\right]=0\end{aligned}\tag{22}$$

The introduction of the following new normalized parameters and variables facilitates the non-dimensionalization of the system

$$\begin{aligned}\tau=\omega_pt,\ V=\frac{v}{D},A=\frac{a}{D},\lambda=\frac{C_B}{2m_p\omega_p},k_{n1}=\frac{k_1}{m_p\omega_p^2},k_{n2}=\frac{k_2D^2}{m_p\omega_p^2},k_{n3}\\=\frac{k_3D^4}{m_p\omega_p^2},k_{n4}=\frac{k_4D^6}{m_p\omega_p^2},\mu=\frac{m_B}{m_p},\beta=\frac{b}{m_p},s=\frac{S}{D}\end{aligned}\tag{23}$$

where τ is the dimensionless time; V is the displacement of nonlinear oscillator normalized by a characteristic length D, which is the height of beam structure in this study; A is the pth-order modal coordinate normalized by a characteristic length D; λ is damping coefficient of BNESI; k_{n1}, k_{n2}, k_{n3}, and k_{n4} are normalized stiffness coefficients of the BNESI; μ is the mass ratio of the BNESI; β is the inertance ratio of the BNESI; s is the spring offset distance normalized by a characteristic length D.

Those normalized parameters bring Eq. (21) and Eq. (22) into the following non-dimensional form:

$$\begin{aligned}&A_p(\tau)+2\xi_p\dot{A}_p(\tau)+\ddot{A}_p(\tau)+\{k_{n1}\left[V(\tau)-\phi_p(d)A_p(\tau)\right.\\&\left.-s\right]-k_{n2}\left[V(\tau)-\phi_p(d)A_p(\tau)-s\right]^3+k_{n3}\left[V(\tau)-\phi_p(d)A_p(\tau)-s\right]^5\\&-k_{n4}\left[V(\tau)-\phi_p(d)A_p(\tau)-s\right]^7-2\lambda\left[\dot{V}(\tau)-\phi_p(d)\dot{A}_p(\tau)\right]\}\phi_p(d)\\&-\beta\left[\ddot{V}(\tau)-\phi_p(e)\ddot{A}_p(\tau)\right]\phi_p(e)=Q(\tau)\end{aligned}\tag{24}$$

$$\begin{aligned}&\varepsilon\ddot{v}(\tau)-k_{n1}\left[V(\tau)-\phi_p(d)A_p(\tau)-s\right]+k_{n2}\left[V(\tau)-\phi_p(d)A_p(\tau)-s\right]^3\\&-k_{n3}\left[V(\tau)-\phi_p(d)A_p(\tau)-s\right]^5+k_{n4}\left[V(\tau)-\phi_p(d)A_p(\tau)-s\right]^7\\&+2\lambda\left[\dot{V}(\tau)-\phi_p(d)\dot{A}_p(\tau)\right]+\beta\left[\ddot{V}(\tau)-\phi_p(e)\ddot{A}_p(\tau)\right]=0\end{aligned}\tag{25}$$

In these two non-dimensional partial differential equations, the beam coupled with the BNESI system can be simplified to the beam coupled with the CNESI system, if $k_{n1}=k_{n3}=k_{n4}=s=0$. Also, if $k_{n2}=k_{n3}=k_{n4}=s=0$, the beam-BNESI system can be simplified to beam coupled with TMDI system. Hence, the modeling established in this study is universal and strongly nonlinear. Given the strong nonlinear characteristics of this model, we use the Runge-Kutta numerical integration algorithm with adaptive time step to solve the partial differential equation, to ensure accurate obtaining of dynamic response results.

3. VIV mitigation characteristic of BNESI

3.1. Reference bridge and vortex-excited load

The vortex-induced vibration (VIV) commonly occurs in high and long structures due to the vortex shedding frequency of flow approaching the natural frequency of such flexible structures. For the beam structure, such as a long-span bridge, the VIV is frequently observed. Even more important, as the span of the bridge increases, the modal frequencies of the long-span bridge become closer, which results in multimodal VIV (e.g., Xihoumen bridge (Li et al., 2011, 2014). Hence, the exploration of multimodal VIV mitigation for flexible structures is critically important. To examine the VIV responses and their mitigation strategies, this study utilizes a standard six-span continuous bridge structure with a single span length of $L_1=150m$. The unit length mass of the beam structure is $m_s=19800\ kg/m$, and the bending stiffness is $EI=7.6\times10^{11}N\bullet m^2$. The height of the beam deck is $D=4.5m$, and the modal damping ratio is 0.1%. Fig. 3 illustrates the schematic diagram of a continuous beam bridge and the first six mode shapes of this continuous bridge, while Table 1 presents the modal mass and its corresponding modal angular frequency for each mode.

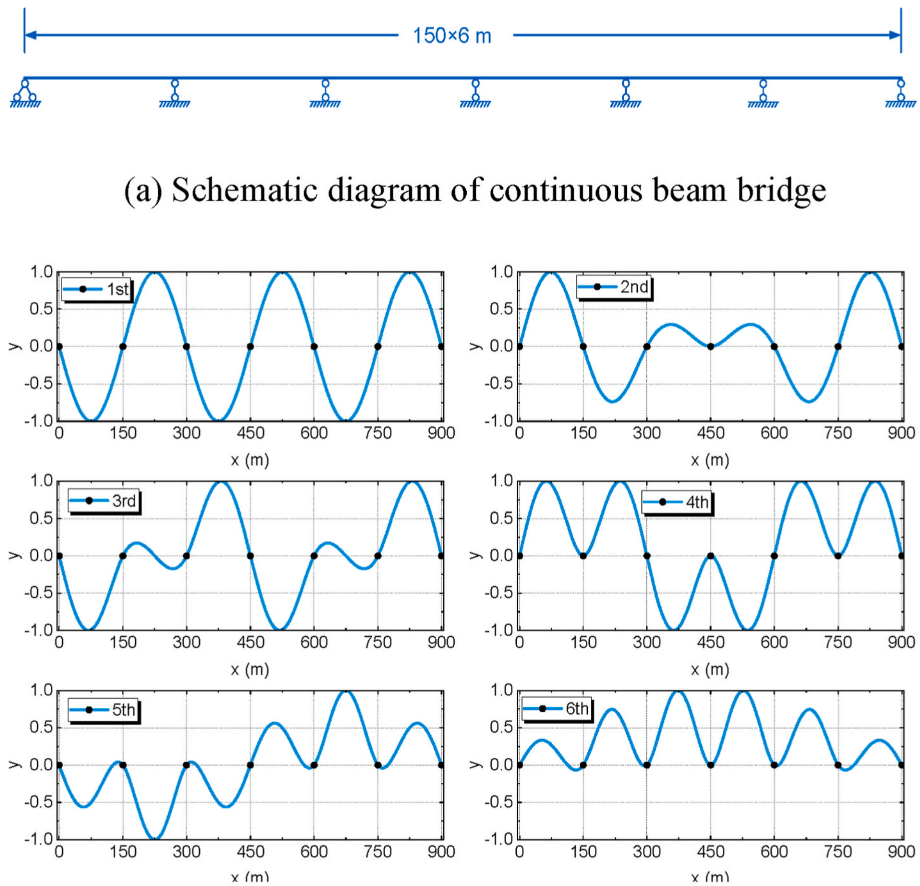
The Scanlan nonlinear vortex-excited force (VEF) model with self-limiting characteristics is employed to characterize external excitations on the bridge (Ehsan and Scanlan, 1990; Yu et al., 2022b; Zhang et al., 2020):

$$Q(t)=\rho U^2D\int_0^L\left[Y_1\left(1-\varepsilon\frac{y^2(x,t)}{D^2}\right)\right]\frac{\dot{y}(x,t)}{U}\phi_p(x)dx\tag{26}$$

where ρ is the air density taken as $1.225kg/m^3$; U is wind velocity; Y_1 and ε are aeroelastic damping coefficients, which are determined as 15.7 and 2900, respectively (Yu et al., 2022b). The dimensionless representation of the VEF model is as follows:

$$Q(\tau)=\frac{U_r}{2\pi m_r}\int_0^1\left[Y_1\left(1-\varepsilon\phi_p^2(\chi L)A_p^2(\tau)\right)\right]\phi_p^2(\chi L)\dot{A}_p(\tau)d\chi\tag{27}$$

where χ is spanwise coordinate normalized by length of beam $\chi=\frac{x}{L}$; U_r and m_r are reduced wind speed and mass ratio of modal mass to air fluid, and they are defined as:



(a) Schematic diagram of continuous beam bridge

(b) Normalized modal shapes

Fig. 3. Configuration and dynamic characteristics of reference bridge.

Table 1
Modal parameters of continuous beam bridge (Yu et al., 2022b).

Modal order	Angular frequency (rad/s)	Modal mass (t)
1	2.721	8887.5
2	2.928	4740.4
3	3.487	5722.5
4	4.247	7803.8
5	5.089	4039.8
6	5.837	3977.4

$$U_r = \frac{2\pi U}{\omega_p D}, m_r = \frac{m_p}{\rho D^2 L} \tag{28}$$

The VIV reduction ratio (denoted as R_i) is determined by evaluating the root-mean-square (RMS) value of the dimensionless displacement of the beam structure. This VIV reduction ratio is calculated by complete displacement time histories from zero to steady state displacement, and the nondimensionalized simulation time is $\tau = 500$ to ensure enough time duration. Each modal VIV reduce ratio is defined as

$$R_i = \frac{RMS_u - RMS_c}{RMS_u} \tag{29}$$

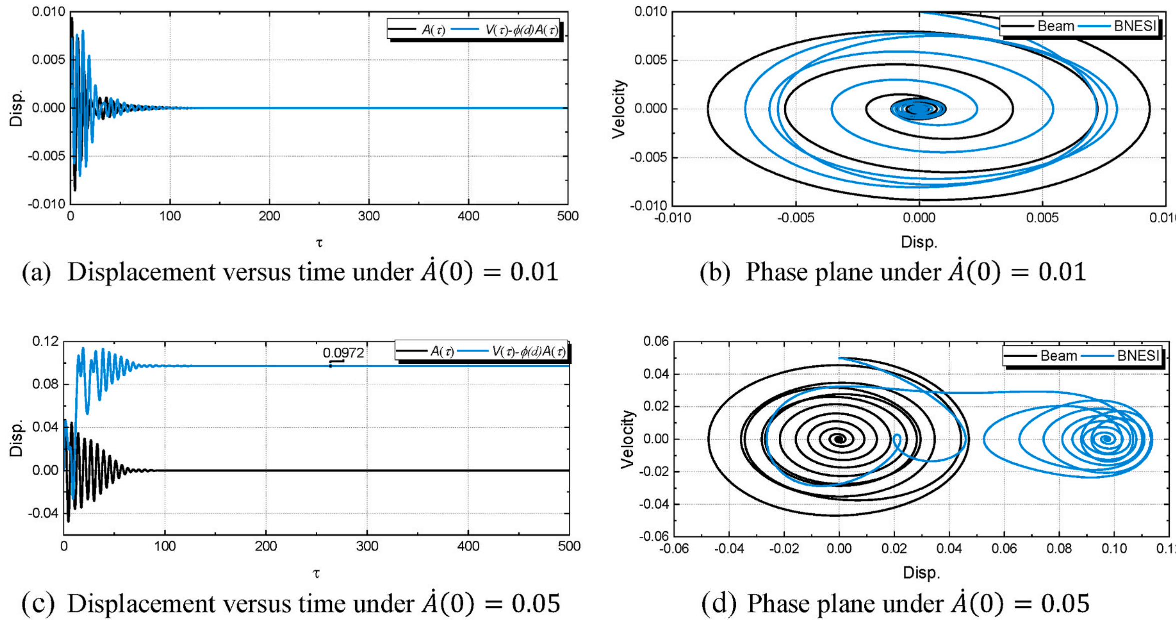
where R_i is modal VIV reduction ratio; RMS_u is the RMS value of the uncontrolled dimensionless displacement of the beam deck; RMS_c is the RMS value of the controlled dimensionless displacement of the beam deck.

3.2. Energy-dependent stationary position

As mentioned above, the BNESI device possesses two stable equilibrium positions and the nonlinear oscillator always oscillates around one of the equilibrium points and ultimately remains stationary at that stable equilibrium point due to energy dissipation. To reveal the dependence relation between the stable position of the BNESI oscillator and input energy, vortex-excited forces with different initial conditions were used to excite the beam deck to obtain its first modal VIV displacement response. In this case, the BNES is connected at the mid-span of the first span, and the two terminals of the inerter are connected with BNES and beam deck respectively, resulting in $\phi_1(d) = 1$ and $\phi_1(e) = 0.6$, as shown in Fig. 4. The inertance ratio β is set as 0.4; the length of the transverse spring is taken as $l = 0.2D$, and l_0 is taken as $0.97l$. The mass ratio μ, stiffness coefficient k_{n2}, and damping coefficient λ of BNESI are selected as 0.97%, 92.15 and 0.06 respectively through the simulated annealing (SA) algorithm to ensure the largest modal VIV reduction ratio. More details of the SA algorithm in parameter optimization in this study will be introduced in the following sections. Only the stiffness coefficient k_{n2} is given here, because according to (4) and (23), once it is determined, other stiffness coefficients are naturally determined.

Fig. 5 presents the time history of dimensionless displacement and the phase plane under VIV with different initial dimensionless velocities of the beam coupled with the BNESI system. Whether under initial low inputted energy ($\dot{A}(0) = 0.01$) or high energy ($\dot{A}(0) = 0.05$), the VIV of beam deck is rapidly mitigated in this coupled system. Interestingly, the stable displacement of a nonlinear oscillator varies at different initial

Fig. 4. Continuous beam structure coupled with BNESI for VIV mitigation.



(a) Displacement versus time under $\dot{A}(0) = 0.01$ (b) Phase plane under $\dot{A}(0) = 0.01$

(c) Displacement versus time under $\dot{A}(0) = 0.05$ (d) Phase plane under $\dot{A}(0) = 0.05$

Fig. 5. Dynamic responses of beam deck and BNESI under VIV.

energy levels. Specifically, at low initial energy, the nonlinear oscillator stabilizes at the left stable equilibrium position ($V(\tau) - \phi(d)A(\tau) = 0$) shown in Fig. 1, and the limit cycle in phase plane is transformed into a sink point (Fig. 5(b)); at high initial energy, the displacement of the nonlinear oscillator increases to oscillate near the right equilibrium point and stabilizes at the right stable equilibrium position ($V(\tau) - \phi(d)A(\tau) = 2s$) (Fig. 5(c)), and the sink point of nonlinear oscillator shifts to the right (Fig. 5(d)). Such a result indicates that the stable equilibrium position of the BNESI oscillator is related to the magnitude of the inputted initial energy.

3.3. Low stiffness characteristics

According to Eqs. (24) and (25), the difference between the beam coupled with the BNESI system and the CNESI coupled to the beam system is reflected in the stiffness coefficient. Thus, there is a difference in the stiffness coefficient required for optimal VIV mitigation of the beam deck between BNESI and CNESI. To identify the stiffness characteristics corresponding to BNESI and CNESI under optimal VIV mitigation, the fixed inertance ratio ($\beta = 0.40$), mass ratio ($\mu = 0.97\%$), and modal value $\phi_e = 0.6$ are used for BNESI and CNESI with the initial condition of $\dot{A}(0) = 0.01$. The SA algorithm is used to search for the optimal cubic stiffness coefficient and damping coefficient of NESI under optimal VIV mitigation for the first mode, and the optimal stiffness and damping of CNESI are determined as $k_n^{opt} = 206.63$ and $c_n^{opt} = 0.20$, which is significantly greater than the cubic stiffness ($k_{n2}^{opt} = 21.45$) and damping ($c_{n2}^{opt} = 0.06$) of BNESI. Fig. 6 shows the dimensionless displacement time histories of the beam deck under optimal coefficients. The displacement of the beam deck coupled with BNESI rapidly decays to 0, while that of the beam deck coupled with CNESI shows a slow displacement attenuation. The VIV reduction ratio (R) of the beam-BNESI system is 0.96, while that of the beam-CNESI system is 0.89, also indicating higher VIV mitigation capacity of BNESI than CNESI.

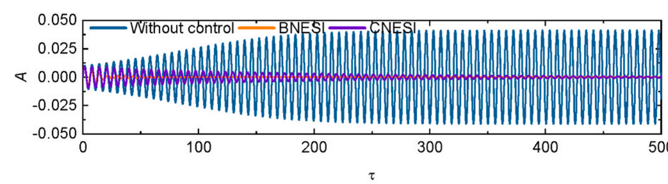


Fig. 6. VIVs of first mode with and without control.

3.4. Multimodal vibration mitigation

To compare the multimodal vibration mitigation efficiency of BNESI and CNESI for the beam deck, the modal VIV reduction ratios of the beam deck under different target design modes are investigated in this section. The optimal modal VIV reduction ratios of the beam deck are determined by the SA algorithm, which is inspired by the annealing process of metal that cools and freezes into a crystalline state (Kirkpatrick et al., 1983). Unlike the gradient-based methods and other deterministic search methods that have the disadvantage of being trapped into local minima, SA's main advantage is its ability to avoid being trapped in local minima. It has been proved that simulated annealing will converge to its global optimality if enough randomness is used in combination with very slow cooling (Yang, 2014). In the SA algorithm, it is important to select a proper initial temperature T_0, which is defined as

$$T_0 \approx -\frac{\max(\Delta f)}{\ln p_0} \tag{30}$$

where max (Δf) is the maximum change of the objective function, which can be determined by $f_{t+1}(\boldsymbol{x}_{t+1}) - f_t(\boldsymbol{x}_t), \boldsymbol{x} = (x_1, \ldots, x_n)$. p_0 is the given probability. The annealing or cooling process is realized by a geometric cooling schedule

$$T(t) = T_0\alpha^t \tag{31}$$

where α is the cooling factor determined as 0.95 in this study, and t is the pseudo time for iterations. For the final temperature, it should be zero in theory so that no worse move can be accepted.

The inertance ratio and installation position are fixed as $\beta = 0.4$ and $\phi_e = 0$; the mass blocks of BNESI and CNESI are installed at the middle of the first span; the mass ratio (μ), damping coefficient (λ), and cubic stiffness coefficient (k_{n2} for BNESI and k_n for CNESI) of the nonlinear oscillator are determined through the SA algorithm. For the beam-BNESI system and beam-CNESI system, the optimization interval of the parameters and objective function are defined as

$$\begin{cases} min\{-R_i\} \\ s.t.\begin{cases} 0.0001 \le \mu \le 0.01 \\ 0.0001 \le \lambda \le 0.5 \\ 0.01 \le k_{n2}(k_n) \le 10000 \end{cases} \end{cases} \tag{32}$$

where R_i is the ith modal VIV reduction ratio.

The optimal coefficients of BNESI and CNESI under different target design modes are listed in Table 2. The optimal stiffness coefficients, damping coefficients, and mass ratio of CNESI are significantly larger than those of BNESI, while the VIV reduction ratio of CNESI is lower than that of BNESI. Figs. 7 and 8 show the modal VIV reduction ratios of the beam-BNESI system and beam-CNESI system under different design modes, respectively. Both BNESI and CNESI are identified to have the ability to mitigate multimodal vibrations, and each design mode vibration mitigation efficiency of BNESI is higher than that of CNESI. For instance, the 1st-order VIV reduction ratio (R_1) of the beam-BNESI system under the 1st-order design mode is greater than that of the beam-NESI system, and the same results can be identified in other design modes. Although the vibration of modes that are far from the target design mode can also be suppressed, their vibration mitigation efficiency is significantly reduced. Compared to other modes, the device with the 4th order mode as the design mode can achieve more efficient multimodal vibration mitigation under optimal parameters. This is because when the 4th mode is selected as the design mode, its frequency is not significantly different from those of other modes. With the robust vibration suppression capability of NESI, modes near the frequency of the 4th mode can achieve comparable VIV mitigation performance. Hence, in multi-mode vibration control, to achieve efficient vibration suppression, it is recommended to select an intermediate mode as the design mode.

4. Sensitive parameter identification

For the beam-BNESI system, the efficiency of multimodal vibration mitigation is controlled by various parameters, including mass ratio of nonlinear oscillator (μ), inertance ratio (β), damping coefficient (λ), nonlinear stiffness coefficient (k_{n2}), spring length ratio ($\alpha = \frac{l_0}{l}$), dimensionless spring offset ($s = \frac{S}{D}$), modal difference ($\Delta_\phi = |\phi_d - \phi_e|$) of the inerter. Multiple parameters are not convenient for practical design, and screening non-sensitive parameters through sensitivity analysis of such parameters is beneficial to reduce the number of parameters used for design. The multimodal vibration control index (*MVCI*) weighted based on the VIV reduction ratio is used to evaluate the multimodal vibration mitigation ability of BNESI under these changing parameters. The importance of each vibrational mode is considered equal; hence they are assigned identical weighting coefficients. The 4th mode is selected as design mode, and the *MVCI* is weighted by VIV reduction ratio as

$$MVCI = \sum_{i=1}^{n} wR_i \tag{33}$$

where n is the number of modes; w is the weighting factor determined as $w = \frac{1}{n}$; R_i, $i = 1, \ldots, 6$, is modal VIV reduction ratio.

When the aforementioned seven parameters of the beam-BNESI system vary, it is difficult to find its optimal *MVCI*. Hence, the SA algorithm is employed to perform global optimization and find the optimal solution. The optimization interval of the parameters and objective function are defined as

$$\begin{cases} min\{-MCVI\} \\ s.t.\begin{cases} 0.0001 \le \mu \le 0.01 \\ 0.1 \le \beta \le 0.6 \\ 0.0001 \le \lambda \le 0.5 \\ 0.01 \le k_{n2} \le 100 \\ 0.8 \le \alpha \le 0.98 \\ 0.01 \le s \le 0.1 \\ 0 \le \Delta_\phi \le 1.0 \end{cases} \end{cases} \tag{34}$$

The optimal value of *MVCI* ($MVCI_{opt}$) is found to be 0.97, along with the corresponding optimal values of seven parameters, as summarized in Table 3. To identify the sensitivity of these parameters, the *MVCI* under parameter variations is plotted as a heatmap, and the ratio of *MVCI* to $MVCI_{opt}$, denoted as η, is used to measure the discrepancy between the results under each parameter and the optimal result. It is worth noting that when exploring the sensitivity of individual parameters, the values of other parameters are fixed at the optimal levels presented in Table 3.

4.1. Effect of mass and inertance ratio

Fig. 9 illustrates the influence of the mass ratio and inertance ratio on η. In comparison to the inertance ratio, it is found that the mass ratio has a weak effect on η, indicating its non-sensitive nature. Conversely, at the same mass ratio level, increasing the inertance ratio significantly enhances *MVCI*, demonstrating the strong sensitivity of *MVCI* to the inertance ratio. Thus, during the BNESI parameter design process, the mass ratio can be selected to a small magnitude to reduce the cost of BNESI, while the inertance ratio needs to be adjusted according to the VIV reduction requirements.

4.2. Effect of damping and stiffness

In Fig. 10, increasing the damping coefficient is found to be

Table 2
Optimal coefficients of BNESI and NESI under different design modes.

Design mode	Mass ratio		Damping coefficient		Cubic stiffness		Optimal R	
	BNESI	CNESI	BNESI	CNESI	BNESI	CNESI	BNESI	CNESI
1	0.45%	0.98%	0.10	0.16	13.74	393.61	0.98	0.97
2	0.42%	1.00%	0.07	0.17	16.72	205.35	0.98	0.97
3	0.29%	0.94%	0.09	0.17	13.85	119.21	0.98	0.97
4	0.42%	1.00%	0.08	0.17	15.47	175.20	0.98	0.97
5	0.87%	0.97%	0.06	0.19	19.00	236.74	0.97	0.93
6	0.70%	0.81%	0.04	0.02	20.87	875.18	0.96	0.50

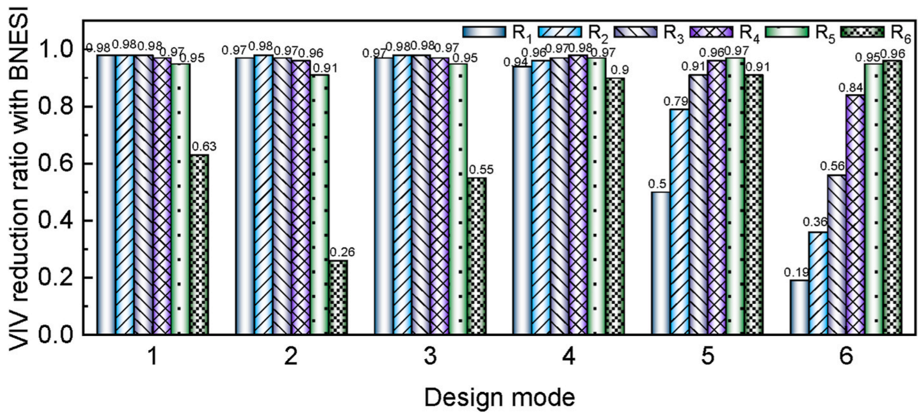


Fig. 7. VIV reduction ratios of beam-BNESI system under different design modes.

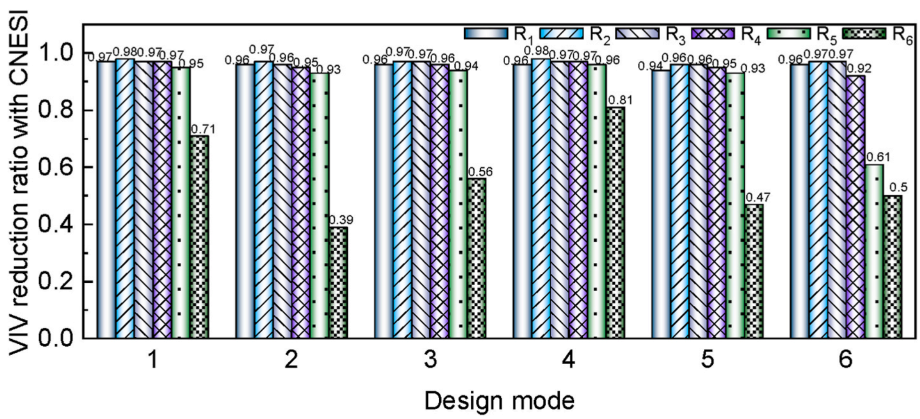


Fig. 8. VIV reduction ratios of beam-CNESI system under different design modes.

Table 3
Optimal parameters of beam-BNESI system under $MVCI_{opt}$.

Parameter	Optimal value
μ	0.04%
β	0.60
λ	0.34
k_{n2}	9.10
α	0.98
s	0.10
Δ_ϕ	1.00

beneficial for enhancing *MVCI*. When the damping coefficient is less than 0.1, the multimodal vibration mitigation performance of VIV is relatively low. At this juncture, a higher stiffness coefficient is required to achieve optimal control performance. Conversely, as the damping coefficient increases, the optimal stiffness coefficient decreases. The conclusion can be drawn that *MVCI* is sensitive to both the damping coefficient and the stiffness coefficient of BNESI.

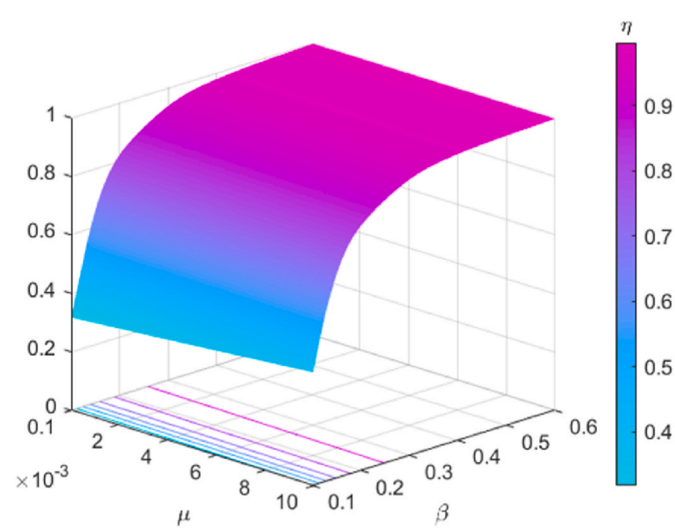


Fig. 9. Influence of mass ratio μ and inertance ratio β on η.

4.3. Effect of spring configuration

Both excessively large and excessively small spring length ratios exhibit improved multi-mode vibration suppression performance compared to intermediate values, as illustrated in Fig. 11. On the contrary, altering the spring offset does not impact the multi-mode control performance. It is noteworthy that although the spring length ratio results in variations in η, the magnitude of these variations is minimal. Hence, both variables can be deemed non-sensitive parameters.

4.4. Effect of inerter location

Fig. 12 depicts the influence of the BNES installation position and inerter installation position on *MVCI*. When BNES and the inerter are installed in the same location, i.e., $\phi_d = \phi_e$, the *MVCI* is minimized, making this installation arrangement the most unfavorable option. However, when BNES and inerter are installed in different locations, the *MVCI* is greater and increases with the increment in the modal

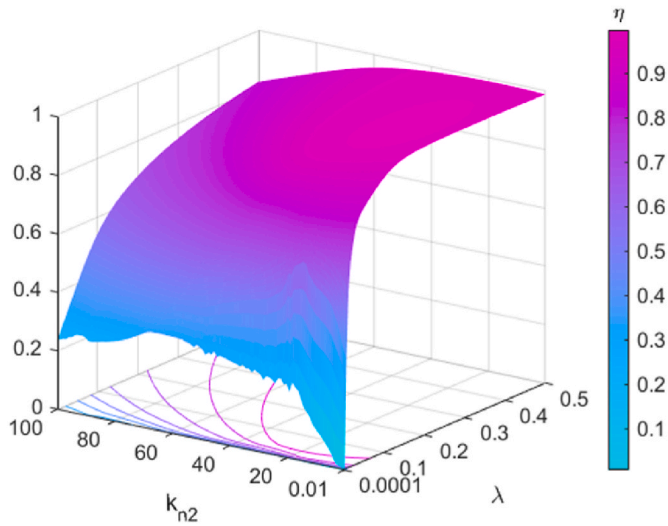


Fig. 10. Influence of damping coefficient λ and stiffness coefficient k_{n2} on η.

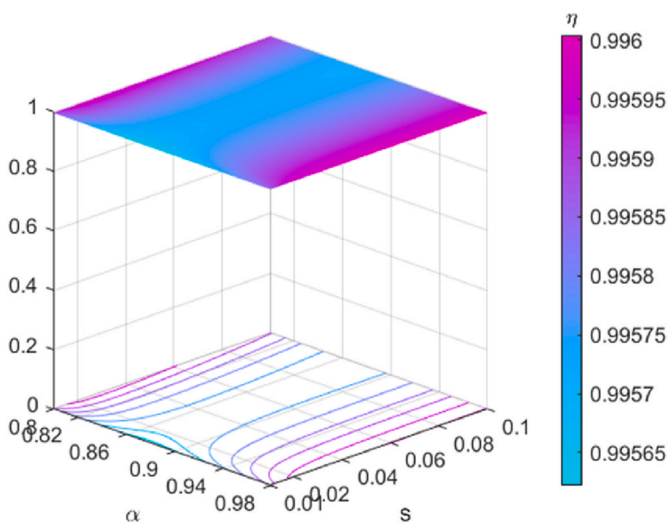


Fig. 11. Influence of spring offset s and spring length ratio α on η

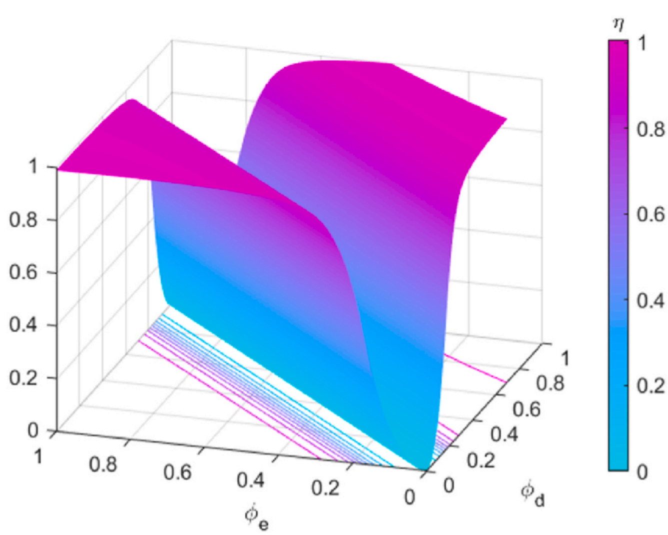


Fig. 12. Influence of installation of BNESI on η

difference. This is attributed to the fact that as the installation spacing increases, the modal vectors of the beam differ more significantly, leading to a more pronounced mass amplification effect caused by inerter, which, in turn, favors the increase of *MVCI*. It can be observed that adopting a modal difference ($\Delta_\phi = |\phi_d - \phi_e|$) greater than 0.4 between BNES and inerter results in high *MVCI*. Hence, the influence of the BNES and inerter installation positions on *MVCI* can be effectively considered equivalent to the influence of the installation spacing between BNES and inerter.

5. Performance-based design approach and model

5.1. Selection of design parameters

For VIV, which does not directly affect the structural load-carrying capacity, designers can perform targeted hierarchical vibration mitigation based on the specific vibration mitigation requirements, known as performance-based VIV control methods. Based on the effectiveness of multimodal vibration mitigation, *MVCI* can be classified into different performance levels as listed in Table 4. The crux of performance-based multimodal vibration mitigation design lies in calculating the corresponding BNESI parameters once the required *MVCI* has been defined. Based on the sensitivity analysis results, the design parameters of BNESI can be reduced to four, i.e., inertance ratio (β), damping coefficient (λ), stiffness coefficient (k_{n2}), and modal difference (Δ_ϕ). The mass ratio, spring length ratio, and spring offset are fixed as 0.0004, 0.98, and 0.1 respectively because of the non-sensitivity on *MVCI*. Given that the optimal parameters result in the lowest cost for BNESI, in performance-based multimodal vibration mitigation design, the SA algorithm is utilized to search for the optimal parameters corresponding to different levels of the *MVCI*. At this situation, the obtained *MVCI* represents the desired *MVCI* (M_d). Hence, the objective function is defined as

$$\begin{cases} min\ (M_d - MCVI)^2 \\ s.t. \begin{cases} 0 \le \beta \le 1.0 \\ 0.0001 \le \lambda \le 0.5 \\ 0.01 \le k_{n2} \le 100 \\ 0 \le \Delta_\phi \le 1.0 \end{cases} \end{cases} \tag{35}$$

5.2. Empirical design models for optimal parameters

The optimal inertance ratio (β) versus *MVCI* is shown in Fig. 13. To achieve an *MVCI* greater than 0.9, the required inertance ratio cannot be less than 0.2. Increasing β is beneficial for increasing *MVCI*, but once the *MVCI* reaches 0.95, further increasing β becomes less effective in increasing *MVCI*. The relationship between inertance ratio and optimal *MVCI* can be defined as

$$\beta = a_0 + a_1 e^{-MVCI/a_2} \tag{36}$$

where a_0, a_1, and a_2 are fixed parameters, and their magnitudes along with their confidence intervals are listed in Table 5.

Fig. 14 illustrates the relationship between the damping coefficient (λ) and the inertance ratio (β) under the target *MVCI*. It can be observed that as the inertance ratio increases, to achieve the desired *MVCI*, the

Table 4
MVCI across various performance levels.

Performance level	Range of index
Perfect	$0.9 \le MVCI$
Excellent	$0.8 \le MVCI < 0.9$
Good	$0.6 \le MVCI < 0.8$
Fair	$0.4 \le MVCI < 0.6$
Poor	$MVCI < 0.4$

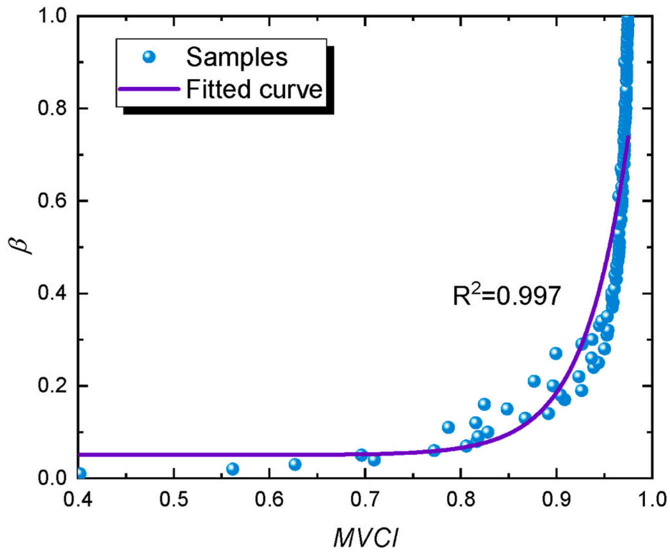


Fig. 13. Inertance ratio versus *MVCI*.

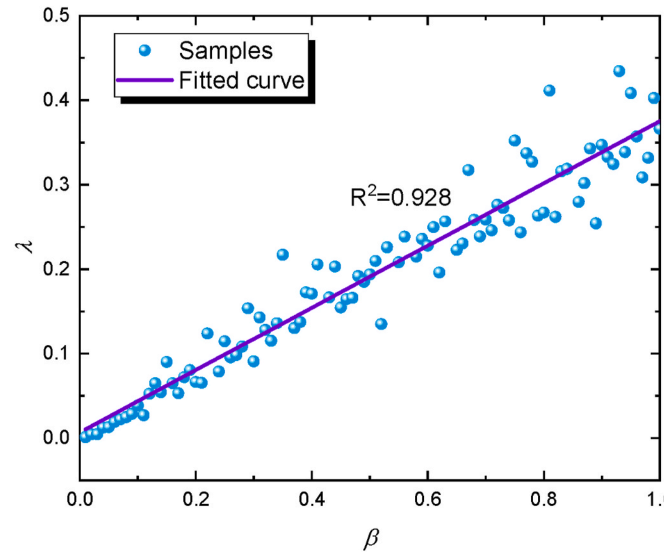


Fig. 14. Damping coefficient versus inertance ratio.

damping coefficient also needs to increase linearly. Hence, the fitting relationship can be described as

$$\lambda = b_0 + b_1\beta \tag{37}$$

where b_0, and b_1 are fixed parameters, and Table 6 lists their magnitudes along with their confidence intervals.

The stiffness coefficient (k_{n2}) versus inertance ratio (β) is depicted in Fig. 15. The stiffness coefficient increases with the increase in inertance ratio, and this increasing trend is particularly significant when β is less than 0.4. The relationship between the stiffness coefficient and the inertance ratio can be represented by the following formula

$$k_{n2} = c_0 + c_1 e^{-\beta/c_2} \tag{38}$$

where c_0, c_1, and c_2 are fixed parameters, which are listed in Table 7.

The modal difference (Δ_ϕ) under the desired *MVCI* shows an increasing trend with an increase in the inertance ratio. However, its variation is significant (see Fig. 16), making it difficult to provide an expression with high fitting accuracy. Hence, to simplify the design of BNESI, the maximum modal difference ($\Delta_{\phi\ max} = 0.96$) is chosen as the recommended value instead of being a parameter that varies with the inertance ratio.

According to the *MVCI* under different performance levels as shown in Table 4 and empirical design models (36), (37), and (38), the performance-based design for the beam-BNESI system can be conducted, and the design process is outlined in Fig. 17. It can be observed that the design parameters for the beam-BNESI system are only three. This is because the nonlinearity oscillator mass ratio ($\mu = 0.0004$), spring length ratio ($\alpha = 0.98$), spring offset ($s = 0.1$), and modal difference ($\Delta_\phi = 0.96$) are fixed parameters in this case due to their non-sensitive characteristics.

Fig. 18 depicts the relationship between the target *MVCI* ($MVCI_t$) and the predicted *MVCI* ($MVCI_p$) obtained using the empirical models employed in this study. It can be observed that when the target *MVCI* value is below 0.5 (corresponding to fair and poor performance levels),

Table 5
Parameter values of the β-*MVCI* model.

Parameter	Magnitude	Confidence interval (95%)
a_0	0.051	0.051 ± 0.006
a_1	4.382e-10	4.382e-10 ± 6.137e-10
a_2	−0.046	−0.046 ± 0.003

Table 6
Parameter values of the $\lambda-\beta$ model.

Parameter	Magnitude	Confidence interval (95%)
b_0	0.007	0.007 ± 0.007
b_1	0.368	0.368 ± 0.011

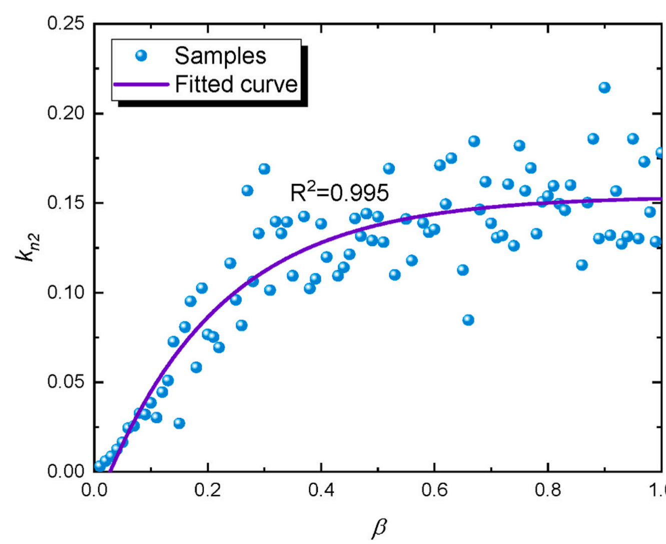


Fig. 15. Stiffness coefficient versus inertance ratio.

Table 7
Parameter values of the $k_{n2}-\beta$ model.

Parameter	Magnitude	Confidence interval (95%)
c_0	0.154	0.154 ± 0.005
c_1	−0.176	−0.176 ± 0.012
c_2	0.210	0.210 ± 0.029

the *MVCI* values calculated using the proposed models of parameters exhibit poor accuracy, resulting in invalid predictive outcomes (red area). Conversely, when the target *MVCI* exceeds 0.5, the *MVCI* values calculated using the proposed models of parameters demonstrate higher accuracy, leading to valid predictive outcomes (blue area). This is

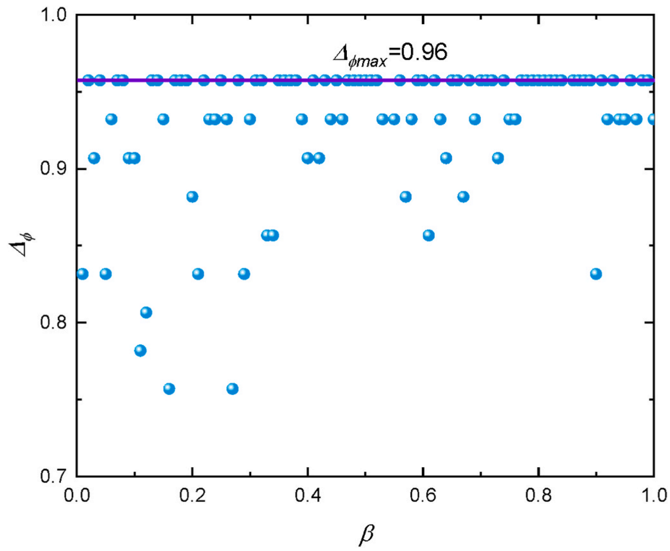


Fig. 16. Modal difference versus inertance ratio.

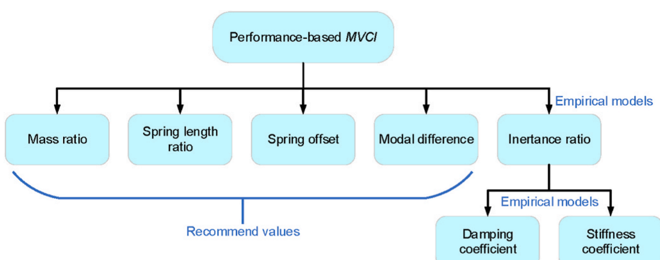


Fig. 17. Design process of BNESI employed for mitigating multi-modal vibration.

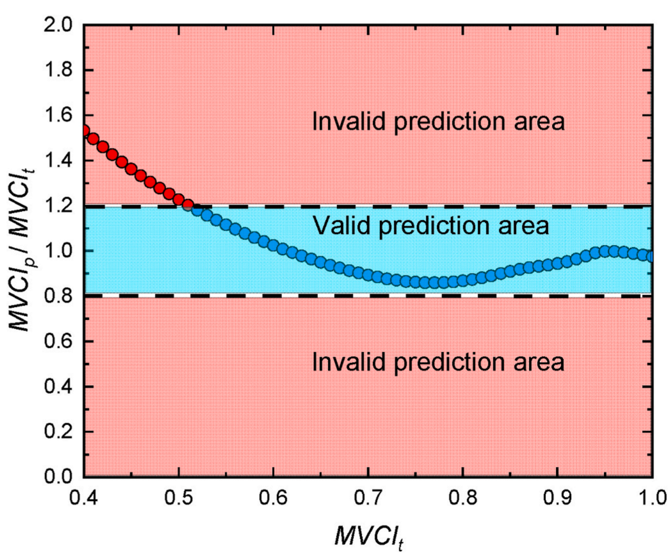


Fig. 18. Predicted *MVCI* versus target *MVCI*.

acceptable because designers typically would not aim to design a poorly performing BNESI; rather, they are more inclined to design a BNESI that can effectively mitigate vibrations. Therefore, when the design performance level is rated as "good" or higher, the resulting *MVCI* obtained from the proposed design method is effective.

6. Conclusions

In this study, the bistable nonlinear energy sink inerter (BNESI) was used to mitigate the multimodal vibration of beam structure. The equilibrium differential equations for the highly nonlinear beam-BNESI system were derived and numerically solved using the simulated annealing (SA) algorithm to determine its optimal parameters and multimodal vibration control index (*MVCI*). The main conclusions are summarized as follows:

(1) This study proposes a strategy employing BENSI for controlling multi-mode VIV in bridge structures. This strategy demonstrates a lower stiffness coefficient and higher efficiency in multi-mode VIV suppression compared to CNESI.
(2) The choice of the design mode has a significant impact on the performance of multi-mode VIV control. This study reveals that selecting an intermediate mode as the design mode yields VIV control parameters that achieve the highest efficiency in multi-mode VIV mitigation.
(3) Parameter sensitivity analysis indicates that the oscillator mass ratio, spring length ratio, and spring offset of the beam-BNESI system are insensitive parameters. During performance-based multi-modal vibration control design, the design parameters of BNESI are inertance ratio, damping coefficient, and nonlinear stiffness coefficient.
(4) The performance-based multimodal vibration mitigation design approach and empirical models proposed in this study exhibit good accuracy when the *MVCI* is greater than 0.5. This is advantageous for guiding the multimodal vibration mitigation design of the beam-BNESI system.

CRediT authorship contribution statement

Ruihong Xie: Writing – original draft, Conceptualization. **Kun Xu:** Methodology. **Houjun Kang:** Validation. **Lin Zhao:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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